

Advanced Microeconomics

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1 Applications of General Equilibrium: Edgeworth Box

1) Consider a pure-exchange, private-ownership economy, consisting in two consumers, denoted by $i = 1, 2$, who trade two commodities, denoted by $l = 1, 2$. Each consumer i is characterized by an endowment vector, $\omega_i \in \mathbb{R}_+^2$, a consumption set, $X_i = \mathbb{R}_+^2$, and regular and continuous preferences, $\succeq_i$ on X_i .

- a) Assuming that the consumers' endowments are $\omega_1 = (1, 2)$ and $\omega_2 = (2, 1)$, respectively, construct the Edgeworth Box relative to economy under consideration. With reference to the same economy, define the following notions: competitive equilibrium, Pareto-efficient allocation, Pareto set, contract curve.

Solution: The Cobb-Douglas Edgeworth Box in this exercise looks as follows:

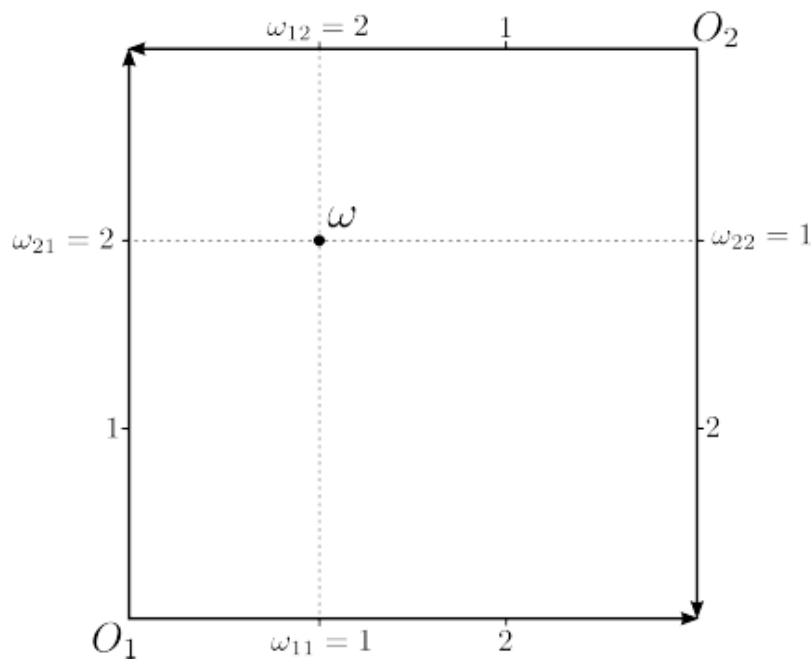


Figure 1: Cobb-Douglas Edgeworth Box

Solution:

The **Pareto Set** is the set of allocations that are Pareto optimal. To find the equation of the interior Pareto Set we must find the locus of points for which the marginal rates of substitution of the two consumers match. Only at those points, in fact, both consumers will be 'happy', not willing to change their behavior. (Note: The utility functions are: $u_i(x_{1i}, x_{2i}) = x_{1i}^{1/3} x_{2i}^{2/3}$)

$$MRS_{12}^1 = \frac{\partial u_1(x_{11}, x_{21})}{\partial(x_{11})} \left(\frac{\partial u_1(x_{11}, x_{21})}{\partial(x_{21})} \right)^{-1} = \frac{1}{2} \frac{x_{21}}{x_{11}}$$

$$MRS_{12}^2 = \frac{\partial u_2(x_{12}, x_{22})}{\partial(x_{12})} \left(\frac{\partial u_2(x_{12}, x_{22})}{\partial(x_{22})} \right)^{-1} = \frac{1}{2} \frac{x_{22}}{x_{12}} = \frac{1}{2} \frac{\bar{\omega}_2 - x_{21}}{\bar{\omega}_1 - x_{11}} = \frac{1}{2} \frac{3 - x_{21}}{3 - x_{11}}$$

$$MRS_{12}^1 = MRS_{12}^2 \Rightarrow \frac{1}{2} \frac{x_{21}}{x_{11}} = \frac{1}{2} \frac{3 - x_{21}}{3 - x_{11}}$$

Therefore the Pareto Set equation is $x_{21} = x_{11}$, i.e. $x_{22} = x_{12}$

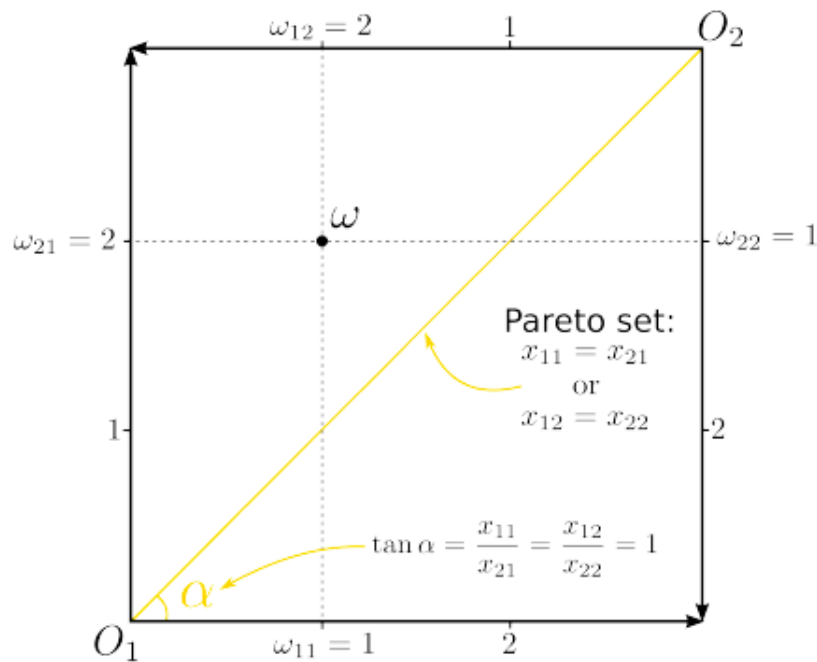


Figure 2: Pareto Set in Edgeworth Box

Solution: The contract curve is the portion of the pareto set for which consumers are at least as well off as by staying with their endowment. We have hence to put into a system the indifference curve passing by the endowment point and the pareto set. By definition, a curve of indifference is the locus of points with the same level of utility. We will hence calculate the level of utility at the endowment allocation, and combine it with the pareto set equation:

$$U_1(\omega_{11}, \omega_{21}) = (1)^{\frac{1}{3}}(2)^{\frac{2}{3}} = 2^{\frac{2}{3}}$$

$$U_2(\omega_{21}, \omega_{22}) = (2)^{\frac{1}{3}}(1)^{\frac{2}{3}} = 2^{\frac{1}{3}}$$

$$\text{For } i = 1, \begin{cases} x_{11} = x_{22} \\ x_{11}^{\frac{1}{3}} x_{21}^{\frac{2}{3}} = 2^{\frac{2}{3}} \end{cases} \quad \text{For } i = 2, \begin{cases} x_{21} = x_{22} \\ x_{21}^{\frac{1}{3}} x_{22}^{\frac{2}{3}} = 2^{\frac{1}{3}} \end{cases}$$

Which can be solved to give the extremal points of the contract curve:
 $x_{11} = x_{21} = 2^{\frac{2}{3}}$ and $x_{21} = x_{22} = 2^{\frac{1}{3}}$

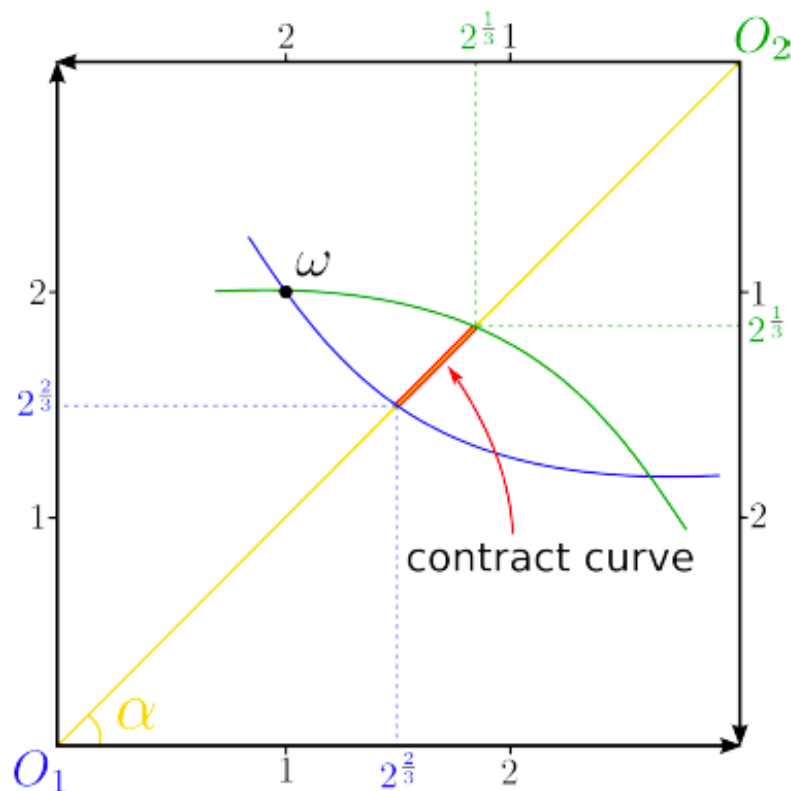


Figure 3: Contract Curve in Edgeworth Box

- b) Find the equation describing the Pareto set (internal solutions); then, taking commodity 1 as the numeraire, hence positing $p_1 = 1$, find the competitive equilibrium allocation and price system; and, finally, draw your results in the Edgeworth Box in each of the following two cases:

- b.1 both consumers' preferences are represented by the same Cobb-Douglas utility function:
 $u_i(x_{1i}, x_{2i}) = x_{1i}^{1/3} x_{2i}^{2/3}$

Solution: Let's impose $p_1 \equiv 1$ and $p_2 \equiv p$. Given these prices and endowments, wealth for the two consumer is: $w_1 = p_1 \omega_{11} + p_2 \omega_{21} = 1 + 2p$; $w_2 = p_1 \omega_{21} + p_2 \omega_{22} = 2 + p$.

Each consumer solves a maximization problem given his budget constraint. From the maximization we get each consumer demand function for each good. Then, by using the fact that $x_{l1} + x_{l2} = \bar{\omega}_l, \forall l = 1, 2$, we can close the system finding:

the vector of equilibrium prices $p^* = (1, p_2^*)$;

the equilibrium allocation x^* .

The max problem of player 1 is:

$$\max : x_{11}^{\frac{1}{3}} x_{21}^{\frac{2}{3}}, \text{ s.t. } x_{11} + p x_{21} = 1 + 2p$$

To reuse previous calculations, remember $MRS_{12}^i = \frac{p_1}{p_2}$.

The solution of the two maximization problems gives us the four demand functions:

$$\begin{aligned} x_{11}(1, p) &= \frac{1}{3}(1 + 2p) \\ x_{21}(1, p) &= \frac{2}{3} \left(\frac{1 + 2p}{p} \right) \\ x_{12}(1, p) &= \frac{1}{3}(2 + p) \\ x_{22}(1, p) &= \frac{2}{3} \left(\frac{2 + p}{p} \right) \end{aligned}$$

These demand functions can be summed over each good to get the total demand for each good. Then, by imposing $x_{l1} + x_{l2} = \bar{\omega}_l, \forall l = 1, 2$ we find prices p^* :

$$\begin{aligned} x_1 &= x_{11}(1, p) + x_{12}(1, p) = 1 + p \\ x_2 &= x_{21}(1, p) + x_{22}(1, p) = 2 \frac{1 + p}{p} \end{aligned}$$

Then, by imposing the condition we get:

$$\begin{aligned} x_1 &= 3 \Rightarrow 1 + p = 3 \Rightarrow p^* = 2 \\ x_2 &= 3 \Rightarrow 2 \frac{1 + p}{p} = 3 \Rightarrow p^* = 2 \end{aligned}$$

By plugging this into the demand of each consumer, we get:

$$\begin{aligned} x_{11}^* &= x_{21}^* = \frac{5}{3} \\ x_{12}^* &= x_{22}^* = \frac{4}{3} \end{aligned}$$

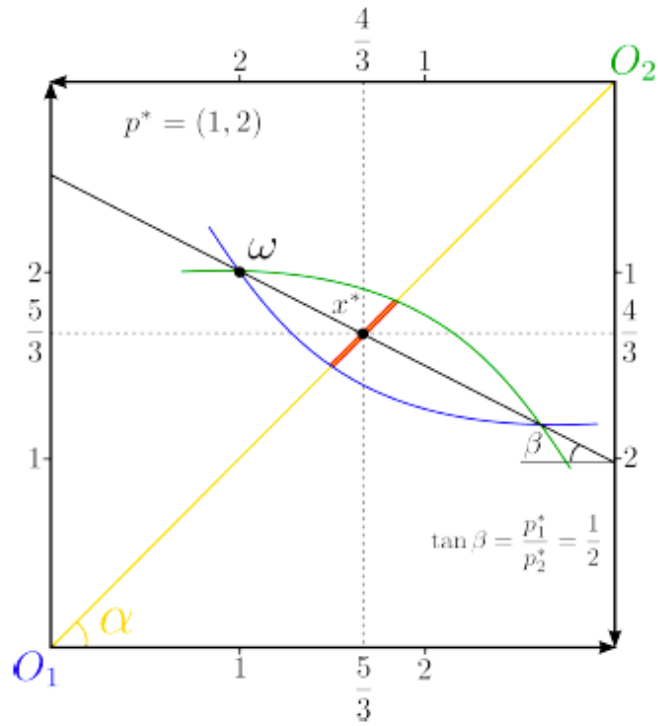


Figure 4: Cobb-Douglas Edgeworth Box Equilibrium (b1)

b.2 the two consumers' preferences are respectively represented by the following quasi-linear utility functions: $u_1(x_{11}, x_{21}) = x_{11} + \ln x_{21}$; $u_2(x_{12}, x_{22}) = x_{12} + 2 \ln x_{22}$

Solution: The solution strategy for quasi-linear preferences is the same as the strategy for the Cobb-Douglas case:

Pareto Set: $x_{21} = 1$

Contract Curve: $x \in \text{Pareto Set s.t. } 1 + \ln 2 \leq x_{11} \leq 1 + 2 \ln 2$

Demand: $x_2 : x_{21}(p, \omega) = \frac{p_1}{p_2}$; $x_{22}(p, \omega) = \frac{2p_1}{p_2}$

Equilibrium prices: $p^* = (1, 1)$

Equilibrium Allocation: $x^* = ((2, 1), (1, 2))$

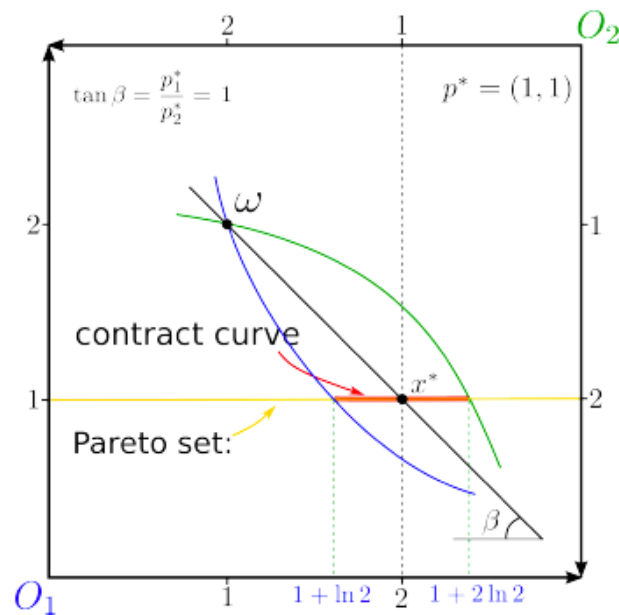


Figure 5: Quasi-Linear Edgeworth Box Equilibrium (b2)

- c) Explain in which of the above two cases the preferences are homothetic and in which case, instead, the preferences are such as to rule out the wealth effects usually associated with price changes. How such peculiar properties of consumers' preferences affect the Pareto set and the shadow prices, i.e., the prices implicit in the Pareto-efficient allocations?

Solution: A preference relation is homothetic if indifference sets are related by proportional expansion along rays. Take the $MRS_{12}^1 = \frac{1}{2} \frac{x_{21}}{x_{11}}$, and take a $\alpha > 0$. Take a ray starting from the origin, equation $x_{21} = \alpha x_{11}$. Therefore $\frac{1}{2} \frac{\alpha x_{11}}{x_{11}} = \frac{\alpha}{2}$. This means that the preferences are homothetic. Homotheticity implies that a ray through the origin is also a isocline. Since we have seen that in this case, with identical preferences for the two consumers, the Pareto set is the diagonal of the Box, it is a ray through the origin for both players, and hence it is an isocline. The relative shadow prices are constant, at $\frac{1}{2}$.

The quasilinear preferences, instead, rule out wealth effects. This implies that the contract curve is horizontal since there is no wealth effect for good 2. To check this, just have a look at the Walrasian demand for good two: it does not depend on wealth. The Shadow prices implicit in the Pareto Set are constant (equal to one in this case).